دستورات لازم برای حل سوالات و ترسیم نمودارها

کانولوشن دایروی N نقطه سیگنال های X1 و X2 :

```
function y = circonvt(x1,x2,N)
% N-point circular convolution between x1 and x2: (time-domain)
% ------------------------------------------------------------
% [y] = circonvt(x1,x2,N)
%  y = output sequence containing the circular convolution
% x1 = input sequence of length N1 <= N
% x2 = input sequence of length N2 <= N
%  N = size of circular buffer
%  Method: y(n) = sum (x1(m)*x2((n-m) mod N))
% Check for length of x1
if length(x1) > N
        error('N must be >= the length of x1')
end
```

```
% Check for length of x2
if length(x2) > N
        error('N must be >= the length of x2')
end
x1=[x1 zeros(1,N-length(x1))];
x2=[x2 zeros(1,N-length(x2))];
m = [0:1:N-1];  x2 = x2(mod(-m,N)+1);  H = zeros(N,N);
for n = 1:1:N
H(n,:) = cirshftt(x2,n-1,N);
end
y = x1*conj(H');
```

مثال:

```
>> x1 = [1,2,2]; x2 = [1,2,3,4];  y = circonvt(x1, x2, 5)
y =
     9     4     9    14    14
```